

Video Scene Graph Generation with Spatial-Temporal Knowledge

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ABSTRACT

Various video understanding tasks have been extensively explored in the multimedia community, among which the video scene graph generation (VidSGG) task is more challenging since it requires identifying objects in comprehensive scenes and deducing their relationships. Existing methods for this task generally aggregate object-level visual information from both spatial and temporal perspectives to better learn powerful relationship representations. However, these leading techniques merely implicitly model the spatial-temporal context, which may lead to ambiguous predicate predictions when visual relations vary frequently. In this work, I propose incorporating spatial-temporal knowledge into relation representation learning to effectively constrain the spatial prediction space within each image and sequential variation across temporal frames. To this end, I design a novel spatial-temporal knowledge-embedded transformer (STKET) that incorporates the prior spatial-temporal knowledge into the multi-head cross-attention mechanism to learn more representative relationship representations. Extensive experiments conducted on Action Genome demonstrate the effectiveness of the proposed STKET.

CCS CONCEPTS

• Computing methodologies → Scene understanding.

KEYWORDS

Video Understanding, Vision and Language, Dynamic Scene Graph Generation

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1 INTRODUCTION

Over the past decades, the multimedia community has extensively explored numerous video understanding tasks, such as video action analysis [9, 25], video captioning [10, 19], and video question answering [15, 27]. Among these tasks, the video scene graph generation (VidSGG) task is more challenging because it requires localizing each semantic object scattered throughout the image and

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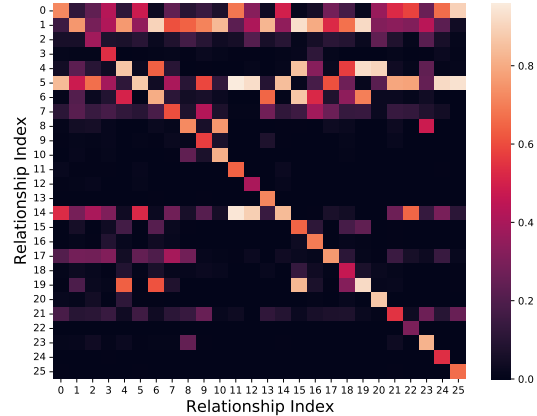


Figure 1: Temporal transition matrix of all relationships on Action Genome [11]. Each row denotes the distribution of transition probabilities after the corresponding relationship exists.

investigating their temporal motion and interactions. With its capacity to depict the semantics of visual content within a holistic scene, VidSGG can underpin plenty of downstream applications of visual content understanding ranging from facial expression recognition [4, 21, 26] and multi-label recognition [2, 3, 20, 22] to recommendation systems [1, 8, 31] and content-based image retrieval [7, 12, 13, 30]. Thus, the task of generating scene graphs in videos has been garnering increasing attention in recent years.

Significant strides have been made in current research by aggregating object-level visual information from spatial and temporal aspects to enhance relationship representation learning [6, 16]. Nonetheless, these state-of-the-art methods may predict ambiguous predicates due to objects' diverse appearances and temporal interactions. Fortunately, there are strong spatial co-occurrence correlations between subject-object pair and their relationships [28]. More specifically, the distribution of relations is highly skewed when the corresponding object categories are given, but not vice versa. Furthermore, I find that there are strong temporal transition correlations between subject-object pairs and their relationships, as depicted in Figure 1. Inspired by these insights, I suggest incorporating spatial-temporal knowledge to effectively constrain the spatial prediction space within each image and sequential variation space across temporal frames.

In this work, I present a novel Spatial-Temporal Knowledge-Embedded Transformer (STKET), which employs multi-head cross-attention layers to incorporate these correlations to guide aggregating spatial-temporal contextual information, enhancing the learning of more representative relationship representations.

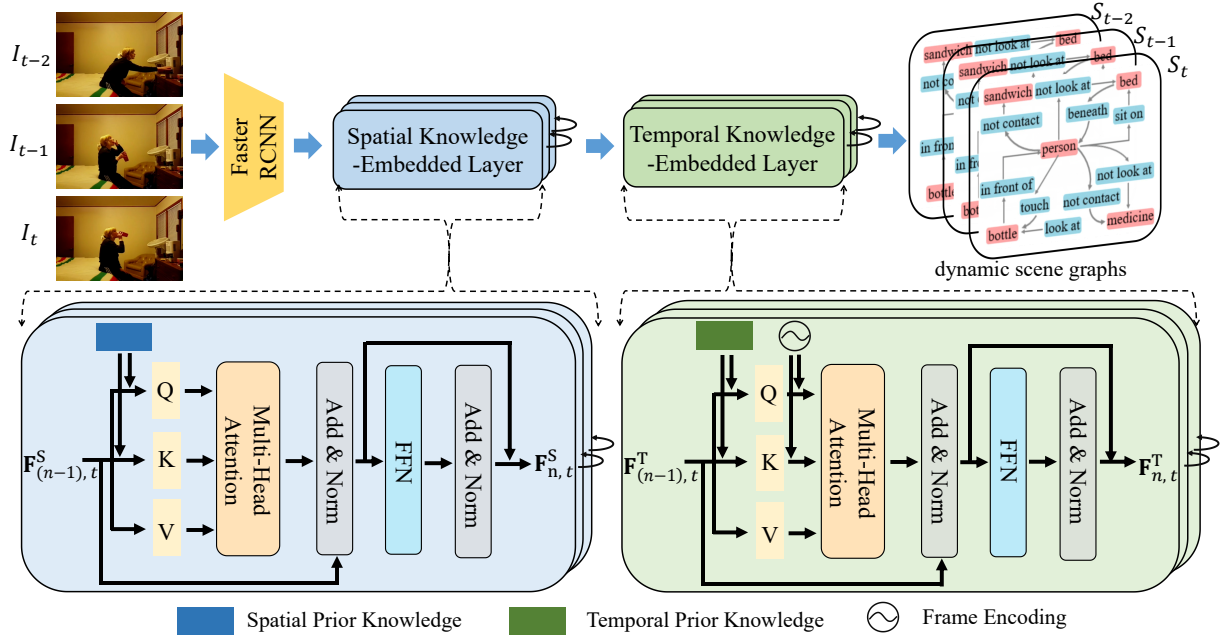


Figure 2: An overall illustration of the proposed STKET framework. It initializes the spatial co-occurrence and temporal transition correlations via statistical matrices from the training set and then embeds them into learnable knowledge representations. Then, it exploits spatial and temporal knowledge-embedded layers to incorporate the spatial co-occurrence and temporal transition correlations into the multi-head cross-attention mechanism to learn spatial- and temporal-embedded representations.

2 METHOD

2.1 Knowledge Representation Learning

Spatial Prior Knowledge. Between different object category pairs, a large difference exists in the spatial co-occurrence correlation of its relationships. To account for this, we construct a spatial co-occurrence matrix $E^{i,j} \in \mathbb{R}^C$ for i -th object category and j -th object category by measuring the frequency of each predicate in the relation set of this pair. C is the total number of types of relationships. For example, let us consider the object category pair of *person* and *cup* (assume their indices are i and j , respectively). At each frame, if both two objects are present, the total number of co-occurrence of this pair $N^{i,j}$ is incremented by 1. Next, if their predicates contain *holding* (assume the index is x), then $e_x^{i,j}$ is incremented by 1. Finally, between *person* and *cup*, the spatial co-occurrence probability of each predicate is calculated as:

$$e_x^{i,j} = e_x^{i,j} / N^{i,j}, \quad x \in \{1, \dots, C\}. \quad (1)$$

To distill spatial co-occurrence correlations, the model learns the spatial knowledge embedding based on the spatial co-occurrence matrix. Specifically, the model generates the corresponding spatial knowledge embedding s_t^k for k -th relationship representation in the frame I_t , the generating process can be formulated as

$$s_t^k = f_{spa}(E^{s(k,t),o(k,t)}), \quad (2)$$

where $s(k, t)$ and $o(k, t)$ denote the category of subject and object of k -th relationship representation in the frame I_t , $f_{spa}(\cdot)$ is implemented by four fully-connected layers which map a C -dimension

vector to a 1936-dimension vector. For brevity, we denote the corresponding spatial knowledge embeddings for all object pairs in the frame I_t by $S_t = \{s_t^1, \dots, s_t^{K(t)}\}$.

Since the distribution of real-world relationships is seriously unbalanced, directly utilizing these spatial knowledge embeddings may degrade the model performance on the less frequent relationships. Thus, we train these embeddings to predict predicates by using the binary cross-entropy loss as the objective function:

$$\mathcal{L}_{spk} = \sum_{t=1}^T \sum_{k=1}^{K(t)} \ell(f(s_t^k), y_t^k), \quad (3)$$

where $\ell(\cdot, \cdot)$ denote the binary cross-entropy loss function, $f(\cdot)$ denote the predicate classifier, y_t^k denote ground-truth predicate labels of k -th relationships in the frame I_t .

Temporal Prior Knowledge. In daily life, the interaction between people and objects is characterized by temporal transitions. To identify relationships at different stages, we construct a temporal transition matrix $\hat{E}^{i,j} \in \mathbb{R}^{C \times C}$ for i -th object category and j -th object category. C is the total number of types of relationships. For instance, let us consider the object category pair of *person* and *cup* as an example (assume the indices are i and j , respectively). If the contacting predicate between *person* and *cup* in the previous frame is *holding* (assume the index is x) and in the current frame is *drinking* (assume the index is y), then $\hat{E}_{x,y}^{i,j}$ is incremented by 1. Finally, between *person* and *cup*, the temporal transition probability

of predicates is calculated as:

$$\hat{e}_{x,y}^{i,j} = \hat{e}_{x,y}^{i,j} / e_x^{i,j}, \quad x, y \in \{1, \dots, C\}. \quad (4)$$

To explore temporal transition correlations, the model learns the corresponding temporal knowledge embedding based on the temporal transition matrix. Specifically, given the predicted object labels and relation labels, the model generates the corresponding temporal knowledge embedding $\mathbf{t}_t^k$ for k -th relationship representation in the frame I_t , the generating process can be formulated as

$$\mathbf{t}_t^k = f_{tem}(\hat{E}_{r(k,t)}^{s(k,t),o(k,t)}), \quad (5)$$

where $s(k, t)$ and $o(k, t)$ denote the category of subject and object of k -th relationship representation in the frame I_t , $r(k, t)$ denotes the predicate of k -th relationship representation in the frame I_t , $f_{tem}(\cdot)$ is implemented by four fully-connected layers which map a C -dimension vector to a 1936-dimension vector. To be specific, we utilize the spatial contextualized representation to predict the relation labels coarsely. For brevity, we denote the corresponding temporal knowledge embeddings for all object pairs in the frame I_t by $\mathbf{T}_t = \{\mathbf{t}_t^1, \dots, \mathbf{t}_t^{K(t)}\}$.

Since there exists complicated predicate transition for different object category pairs, the temporal knowledge embedding may contain inaccurate temporal correlation which may result in sub-optimal performance. Therefore, we train these embeddings to predict predicates at the next frame, by using the binary cross-entropy loss as the objective function:

$$\mathcal{L}_{tpk} = \sum_{t=1}^{T-1} \sum_{i=1}^{K(t)} \ell(f(\mathbf{t}_t^i), \mathbf{y}_{t+1}^i), \quad (6)$$

where we assume k -th relationship representation in the frame I_t and frame I_{t+1} belong the same subject-object pair for easily understanding.

2.2 Knowledge-Embedded Attention Layer

Between object pairs and their relationships, there are apparent spatial co-occurrence correlations within each image and strong temporal transition correlations across different images. Thus, we propose incorporating spatial-temporal knowledge into the multi-head cross-attention mechanism to learn spatial- and temporal-embedded representations. Specifically, we design spatial and temporal knowledge-embedded layers that explore the interaction between visual representation and spatial-temporal knowledge fully. **Exploring Spatial Context.** As shown in Figure 2, we first employ the spatial knowledge-embedded layers (SKEL) to explore spatial co-occurrence correlations within each image. Specifically, we take all relationship representations of the current frame I_t as the input, i.e., $\mathbf{F}_{0,t}^S = \mathbf{X}_t = \{\mathbf{x}_t^1, \dots, \mathbf{x}_t^{K(t)}\}$. Then, the SKEL incorporates corresponding spatial knowledge embeddings in the keys and queries to fuse the information from the relationship representation and its corresponding spatial prior. Since we stack N_s identical spatial knowledge-embedded layers, the input of the n -th layer is the output of $(n-1)$ -th layer. Thus, the queries $\mathbf{Q}$, keys $\mathbf{K}$, values $\mathbf{V}$ and

output of the n -th SKEL is presented as:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{F}_{(n-1),t}^S + \mathbf{S}_t, \quad (7)$$

$$\mathbf{K} = \mathbf{W}_K \mathbf{F}_{(n-1),t}^S + \mathbf{S}_t, \quad (8)$$

$$\mathbf{V} = \mathbf{W}_V \mathbf{F}_{(n-1),t}^S, \quad (9)$$

$$\mathbf{F}_{n,t}^S = \text{Att}_{spa}(\mathbf{Q}, \mathbf{K}, \mathbf{V}), \quad (10)$$

where the $\mathbf{W}_Q$, $\mathbf{W}_K$ and $\mathbf{W}_V$ denote the linear transformations, the $\text{Att}_{spa}(\cdot)$ denote cross-attention layer in the SKEL. For simplicity, we remove the subscript n and denote the final output of the SKEL as $\mathbf{F}_t^S$.

Parsing Temporal Correlation. As claimed in prior works [6, 16], the dynamic visual relation can be easily recognized with the given temporal information in the previous frame. Thus, we design the temporal knowledge-embedded layer (TKEL) to explore the temporal correlation in the current frame I_t and the previous frame I_{t-1} . At first, we adopt a sliding window over the sequence of spatial contextualized representation $[\mathbf{F}_1^S, \dots, \mathbf{F}_T^S]$, and the input of the frame I_t in TKEL is presented as:

$$\mathbf{F}_{0,t}^T = [\mathbf{F}_{t-1}^S, \mathbf{F}_t^S], \quad t \in \{2, \dots, T\}. \quad (11)$$

Then, TKEL incorporates corresponding spatial and temporal knowledge embeddings in the keys and queries to fuse the information from the relationship representation and its corresponding spatial and temporal prior. Since we stack N_t identical temporal knowledge-embedded layers, the input of the n -th layer is the output of $(n-1)$ -th layer. Thus, the queries $\mathbf{Q}$, keys $\mathbf{K}$, values $\mathbf{V}$ and output of the n -th TKEL is presented as:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{F}_{(n-1),t}^T + [\mathbf{T}_{t-1}, \mathbf{S}_t] + \mathbf{E}_f, \quad (12)$$

$$\mathbf{K} = \mathbf{W}_K \mathbf{F}_{(n-1),t}^T + [\mathbf{T}_{t-1}, \mathbf{S}_t] + \mathbf{E}_f, \quad (13)$$

$$\mathbf{V} = \mathbf{W}_V \mathbf{F}_{(n-1),t}^T, \quad (14)$$

$$\mathbf{F}_{n,t}^T = \text{Att}_{tem}(\mathbf{Q}, \mathbf{K}, \mathbf{V}), \quad (15)$$

where $\mathbf{E}_f$ is the learned frame encoding vector, the $\text{Att}_{tem}(\cdot)$ denote cross-attention layer in TKEL and the $\mathbf{W}_Q$, $\mathbf{W}_K$ and $\mathbf{W}_V$ denote the linear transformations. Considering that the relationships in a frame have various representations in different batches, we choose the earliest representation appearing in the sliding window. For simplicity, we remove the subscript n and denote the final output of TKEL as $\mathbf{F}_t^T$.

3 EXPERIMENTS

3.1 Experiment Setting

Task. Following prior arts [5, 6, 24], I evaluate our proposed method and other state-of-the-art methods under three kinds of experiment setups: **Predicate Classification (PredCls)**: predict the predicates of object pairs with given ground truth bounding boxes and category labels. **Scene Graph Classification (SGCls)**: predict both the predicates and the category labels of objects with given ground-truth bounding boxes. **Scene Graph Generation (SGGen)**: simultaneously detects objects appearing in the image and predicts the predicates of each object pair. In the *SGGen*, an object bounding box is considered to be correctly detected only if the predicted object

Method	With Constraint									Average Recall
	Pred Cls			SG Cls			SG Gen			
	R@10	R@20	R@50	R@10	R@20	R@50	R@10	R@20	R@50	
Motif [28]	62.4	65.1	65.1	40.8	41.9	41.9	23.7	31.4	33.3	45.1
KERN [5]	64.1	67.1	67.1	40.7	41.6	41.6	23.2	31.9	34.8	45.8
VCTREE [23]	66.0	69.3	69.3	44.1	45.3	45.3	24.4	32.6	34.7	47.9
ReIDN [29]	66.3	69.5	69.5	44.3	45.4	45.4	24.5	32.8	34.9	48.1
GPS-Net [17]	66.8	69.9	69.9	45.3	46.5	46.5	24.7	33.1	35.1	48.6
STTran [6]	68.6	71.8	71.8	46.4	47.5	47.5	25.2	34.1	37.0	50.0
Ours	70.7	73.8	73.8	47.7	48.8	48.8	25.8	34.3	37.0	51.2

Table 1: Comparison with state-of-the-art image-based scene graph generation methods on Action Genome [11] under the with constraint strategy. The same object detector is used in all baselines for fair comparisons.

Method	Without Constraint									Average Recall
	Pred Cls			SG Cls			SG Gen			
	R@10	R@20	R@50	R@10	R@20	R@50	R@10	R@20	R@50	
Motif [28]	73.4	92.4	99.6	50.4	60.6	64.2	22.8	34.3	46.4	60.5
KERN [5]	73.7	92.3	98.8	50.5	62.4	66.3	23.3	35.2	47.0	61.1
VCTREE [23]	75.5	92.9	99.3	52.4	62.0	65.1	23.9	35.3	46.8	61.5
ReIDN [29]	75.7	93.0	99.0	52.9	62.4	65.1	24.1	35.4	46.8	61.6
GPS-Net [17]	76.2	93.6	99.5	53.6	63.3	66.0	24.4	35.7	47.3	62.2
STTran [6]	77.9	94.2	99.1	54.0	63.7	66.4	24.6	36.2	48.8	62.8
Ours	80.6	95.4	99.3	56.1	64.4	66.3	25.4	37.3	49.0	63.8

Table 2: Comparison with state-of-the-art image-based scene graph generation methods on Action Genome [11] under the without constraint strategy. The same object detector is used in all baselines for fair comparisons.

bounding box has at least 0.5 IoU (Intersection over Union) overlap with the ground-truth object bounding box.

Evaluation Metric. All tasks are evaluated with the Recall@K (short as R@K) metric ($K = [10, 20, 50]$), which measures the ratio of correct instances among the top- K predicted instances with the highest confidence. The Recall@K metric is based on two typical generation strategies. (1) **With Constraint:** Each subject-object pair can only have at most one predicate. (2) **Without Constraint:** Each subject-object pair is allowed to have multiple predicates simultaneously.

3.2 Comparison with State-of-the-arts

As presented in Table 1 and 2, through implicitly exploring temporal dependencies, recent video-based algorithm (i.e., STTran [6]) obtains quite a marginal improvement over the image-based SGG algorithms (i.e., VRD [18], Motif [28], MSDN [14], VCTREE [23], KERN [5], ReIDN [29], GPS-Net [17]), demonstrating the effectiveness of temporal correlations for recognizing dynamic visual relationships.

Through incorporating spatial co-occurrence and temporal transition knowledge, the proposed STKET framework performs best on all kinds of experiment setups under the *with constraint* strategy. Specifically, our proposed STKET framework improves the second-best algorithm by 2.1%/2.0% on *Pred Cls* with R@10/20, 1.3%/1.3% on *SG Cls* with R@10/20 and 0.6%/0.2% on *SG Gen* with R@10/20 under the with constraint strategy.

Moreover, our proposed STKET framework outperforms current state-of-the-art methods under the *without constraint* strategy. As shown in Table 2, its average recall is 63.8%, outperforming the second-best STTran algorithm by 1.0%. Specifically, our proposed STKET framework improves the second-best algorithm by 2.7%/1.2% on *Pred Cls* with R@10/20, 2.1%/1.3% on *SG Cls* with R@10/20 and 0.8%/0.9% on *SG Gen* with R@10/20 under the without constraint strategy.

4 CONCLUSION

In this work, I propose incorporating spatial-temporal knowledge to effectively constrain the spatial prediction space within each image and sequential variation space across temporal frames. To this end, I design a novel spatial-temporal knowledge-embedded transformer (STKET) that introduces multi-head cross-attention layers to incorporate these correlations to guide aggregating spatial-temporal contextual information, enhancing the learning of more representative relationship representations. Extensive experiments illustrate its superiority over current state-of-the-art algorithms.

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